

HASSELBLAD XCD

Field of View Comparator

User Guide

Version 2 · Standalone HTML tool for the Hasselblad X2D / X2D II

Contents

1. Overview.....	3
2. Getting Started.....	3
3. Crop Ratio Selector.....	3
4. Lens Select Panel.....	4
4.1 Series groups.....	4
4.2 Zoom lenses.....	4
4.3 Kit presets.....	4
5. Visualisation Views.....	4
5.1 Angle Wedge.....	4
5.2 Scene Overlay.....	4
5.3 Image Overlay.....	5
6. Computed Optics Table.....	5
7. Tips & Notes.....	5
Appendix A — Full XCD Lens Reference.....	6

1. Overview

This tool models the field of view of every current Hasselblad XCD lens on the X2D / X2D II sensor, and lets you compare, overlay, and preview framing across the full lineup — including zooms at any point in their range, six sensor crop ratios, and two synthetic visualisations plus a real-photo overlay mode.

It is a single self-contained HTML file. No installation, account, or internet connection is required after the first load — open it in any modern desktop or mobile browser.

Figures are geometric approximations from nominal (or OEM-measured effective) focal lengths. Real optics have field curvature and slight focus breathing that this simulator does not model.

2. Getting Started

The tool is arranged in two columns:

- Left — Lens Select panel: crop ratio selector, kit presets, and the full 18-lens picker grouped by series.
- Right — Visualisation panel with three tabs (Angle Wedge, Scene Overlay, Image Overlay), followed by the Computed Optics data table.

Tick any lens checkbox to add it to all three visualisations and the data table simultaneously — selections are shared across every view, so you can switch tabs without losing your comparison set.

3. Crop Ratio Selector

A single dropdown at the top of the Lens Select panel controls the active sensor crop. It affects every view at once — the FOV maths in the data table, the Angle Wedge, the Scene Overlay, and the Image Overlay framelines all recompute against whichever crop is active.

Crop	Effective Sensor	Description
4:3 Native	43.8 × 32.9mm	The full X2D sensor — no crop applied.
3:2 Classic	43.8 × 29.2mm	Familiar 35mm-still aspect; trims top/bottom.
XPan 65:24	43.8 × 16.2mm	Ultra-wide panoramic letterbox, styled on the Hasselblad XPan.
1:1 Square	32.9 × 32.9mm	Square crop; trims the sides evenly.
5:4	41.1 × 32.9mm	Traditional print/fine-art ratio; trims the sides slightly.
16:9	43.8 × 24.6mm	Widescreen video-style crop; trims top/bottom.

Crops are always fitted inside the native 43.8 × 32.9mm sensor area (never larger), matching how in-camera and Phocus crop guides behave.

4. Lens Select Panel

4.1 Series groups

Lenses are grouped into four series, each with its own colour family so overlapping framelines stay legible even with several lenses active:

- V — Versatile: 25V, 38V, 55V, 90V (amber/orange shades)
- P — Portable: 28P, 45P, 75P (blue-violet shades)
- E — Exclusive zooms: 20-35E, 35-100E (cyan shades)
- Original Generation: 21, 30, 45, 65, 80, 90, 120 Macro, 135, 35-75 (mixed distinct hues)

4.2 Zoom lenses

The 20-35E, 35-100E, and 35-75 expose a range slider beneath their row once checked. Drag it to any focal length within the lens's travel — every view updates live, and the data table reflects the exact focal length selected, not just the end points.

4.3 Kit presets

Four one-tap presets are provided above the lens list:

- Japan Trio — 25V / 38V / 90V
- Zoom Core — 25V / 35-100E / 90V
- Full V-Series — 25V / 38V / 55V / 90V
- Wide Compare — 21 / 25V / 28P / 20-35E

Clear All removes every active selection.

5. Visualisation Views

5.1 Angle Wedge

A top-down fan diagram showing the horizontal field of view of every active lens as a wedge radiating from a shared camera position. Wider lenses produce broader wedges; narrower (longer) lenses nest inside them. Each wedge is labelled with its horizontal FOV in degrees, making it easy to see at a glance how much wider one lens is than another.

5.2 Scene Overlay

A synthetic scene showing nested rectangles representing the framing each active lens would capture at a chosen subject distance, plus a 1.75m human figure for scale.

- Subject Distance slider — sets the distance (1–50m) at which framing is calculated.
- Reference Figure toggle — shows or hides the 1.75m scale figure.

Each rectangle is labelled with its lens name and its exact frame width × height in metres at the chosen distance — useful for questions like "at 5m, does the 90V get me head-to-toe or just a torso?"

5.3 Image Overlay

Upload or drag-and-drop your own photograph, and every active lens draws its frameline directly on it.

- Shot At (mm) — the focal length the uploaded photo was actually taken with. This is the reference frame that all lens crops are measured against.
- Auto (Widest) — sets "Shot At" to match your widest currently active lens.
- Mask — darkens everything outside the narrowest (most telephoto) active lens's crop, useful for previewing exactly what a tight crop keeps.
- Clear — removes the loaded image.

Framelines are scaled by the ratio of the reference focal length to each lens's focal length, then fitted to the active crop ratio — so switching crop ratios reshapes every frameline without needing to re-upload.

A sample landscape photo is preloaded by default so the Image Overlay tab has something to show immediately. Uploading or dragging in your own photo replaces it.

6. Computed Optics Table

Beneath the visualisation panel, a live table lists every active lens with:

- Focal length (current position, for zooms) and full-frame-equivalent focal length ($\times 0.79$ crop factor)
- Maximum aperture and front filter thread size
- Horizontal, Vertical, and Diagonal field of view in degrees, computed against the active crop

Field of view is computed as $2 \times \text{atan}(\text{crop_dimension} \div (2 \times \text{focal_length}))$, using the crop's effective width, height, or diagonal as appropriate.

7. Tips & Notes

- Two lenses in the range have OEM-measured effective focal lengths that differ slightly from their marketing names: the 45P measures 46.2mm, and the 35-100E measures 36–97mm rather than 35–100mm. The tool uses these true values for all FOV maths while keeping the familiar marketing names as labels.
- Because framelines in every view are centred and proportional to $1/\text{focal length}$, a narrower lens's frame is always nested exactly inside a wider lens's frame at the same distance.
- The tool works entirely offline once loaded — useful for checking a kit's coverage in the field without a signal.

Appendix A — Full XCD Lens Reference

True focal length reflects OEM-measured effective values where they differ from the marketing name (45P, 35-100E); all others match their nominal name.

Lens	Series	True Focal	Aperture	FF Equiv	Filter
XCD 3.2-4.5/20-35E	E	20-35mm	f/3.2-4.5	16-28mm	77mm
XCD 2.8-4/35-100E	E	36-97mm	f/2.8-4	28-77mm	86mm
XCD 2.5/25V	V	25mm	f/2.5	20mm	72mm
XCD 2.5/38V	V	38mm	f/2.5	30mm	72mm
XCD 2.5/55V	V	55mm	f/2.5	43mm	72mm
XCD 2.5/90V	V	90mm	f/2.5	71mm	72mm
XCD 4/28P	P	28mm	f/4	22mm	72mm
XCD 4/45P	P	46.2mm	f/4	36mm	62mm
XCD 3.4/75P	P	75mm	f/3.4	59mm	72mm
XCD 4/21	O	21mm	f/4	17mm	77mm
XCD 3.5/30	O	30mm	f/3.5	24mm	77mm
XCD 3.5/45	O	45mm	f/3.5	36mm	67mm
XCD 2.8/65	O	65mm	f/2.8	51mm	67mm
XCD 1.9/80	O	80mm	f/1.9	63mm	77mm
XCD 3.2/90	O	90mm	f/3.2	71mm	67mm
XCD 3.5/120 Macro	O	120mm	f/3.5	95mm	77mm
XCD 2.8/135	O	135mm	f/2.8	107mm	77mm
XCD 3.5-4.5/35-75	O	35-75mm	f/3.5-4.5	28-59mm	77mm